



SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

**Adaptation Techniques for using NAS
Methods in the FL Setting**

Max Coetzee





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Titel

Author:	Max Coetzee
Examiner:	Supervisor
Supervisor:	Advisor
Submission Date:	Submission date



I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

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Max Coetzee

Acknowledgments

Abstract

Both *Neural Architecture Search* (NAS) and *Federated Learning* (FL) have made significant progress independently in the past decade. To benefit from the advantages of NAS methods in FL, researchers have started combining them by using NAS in FL [9] [5] [13].

[TODO: overhaul]

Applying techniques from Neural Architecture Search (NAS) to Federated Learning (FL) has been fruitful (remove) in recent years. The combination was identified as a promising research (remove) direction by [11]. It has yielded methods for finding architectures that deal with the challenges imposed by the FL setting.

Research into NAS has grown rapidly [18] since it was popularized by [20]; consequently, literature on its application to FL has grown. The last survey on NAS applied to FL compared approaches of four papers [19]. Since then, we have identified approximately 50 new papers. This motivates a new systematic survey of the landscape to identify progress and gaps in the literature.

In this thesis, we propose a map of the literature landscape based on the FL challenges they address. We achieve this by systematically evaluating the literature and identifying which challenge it solves.

We refer to the FL challenges described in [20], i.e., non-IID data, limited communication, client heterogeneity, privacy of client data, and break them down into smaller subchallenges — each subchallenge being associated with a pattern in the literature. We include personalized FL [17] as an additional subchallenge that was not originally posited, but has since drawn the community’s attention.

We then analyze how the subchallenges are addressed and focus on the contribution of the used NAS method towards overcoming the subchallenge. For each subchallenge, we keep track of the NAS types used (following [18], [2]) and assess whether the underexplored methods are candidates for future research.

Neural Architecture Search

Contents

Acknowledgments	iv
Abstract	v
1 Introduction	1
2 Background and Related Work	4
2.1 Neural Architecture Search	4
2.2 Federated Learning	4
2.3 Federated Neural Architecture Search	4
3 Method	5
3.1 Reviewed Literature	6
4 FedNAS Challenges	7
5 Adaptation Techniques	11
6 Discussion	12
7 Conclusion	13
Abbreviations	14
List of Figures	15
List of Tables	16
Bibliography	17

1 Introduction

Deep learning has seen widespread adoption over the last decade and is being applied to an increasingly diverse set of tasks. Applying deep learning traditionally involves a team of deep learning and domain experts who tailor a neural network architecture to a specific task through a lengthy process of trial and error. To automate this process, researchers invented *Neural Architecture Search* (NAS) [7] methods. NAS methods employ diverse strategies to automatically search for a neural network architecture for a given deep learning task within an architecture search space.

Independently, but in parallel to NAS, researchers developed a machine learning approach called *Federated Learning* (FL) [14], which enables training an ML model on data distributed across a set of clients without sharing their data. The model is distributed from the server to the clients, which train the model on their local data. The clients then send the model weight updates to the server, which aggregates them and redistributes them to the clients. This process is repeated for several rounds. FL enhances the privacy of client data and enables training even when clients' data is too large to transfer effectively to a central training site or cannot be shared due to regulatory or privacy constraints.

When practitioners¹ make use of deep learning in FL, they typically use a predefined architecture. However, clients' data distributions and hardware capabilities are highly heterogeneous in FL. Since practitioners cannot directly observe these client characteristics [9], it is hard to choose a single architecture that generalises well across all clients and that meets inference latency constraints on every client. Using NAS methods in FL provides an alternative: the server can coordinate an architecture search across clients, guided by performance feedback from candidate architectures evaluated directly on clients. Thus, an architecture can be found that is tailored to the clients' characteristics.

Despite the benefit, using NAS in FL is not straightforward. Most NAS methods were developed for a centralised setting and rely on assumptions that often do not hold for FL. These assumption discrepancies create several *challenges* for using NAS in FL. For example, in a centralised setting, practitioners can expect a NAS method to run on worker nodes connected via low-latency, high-bandwidth links. However, in FL, a NAS method may run on clients connected via an unreliable, high-latency, low-bandwidth network. This creates a challenge when transferring the weights of

¹In this thesis, *practitioners* refers to both users and developers of NAS methods for FL.

large candidate models between clients and the server to coordinate an architecture search. Practitioners need specialised techniques to overcome such challenges, such as only transferring encodings of candidate architectures to clients [16].

Many techniques for overcoming such challenges already exist, but practitioners need to comb an increasingly large body of NAS-in-FL literature for suitable techniques (i.e., techniques that overcome the challenges relevant to their targeted FL system characteristics). The relevance of a challenge depends on the extent to which the targeted FL system characteristics violate centralised NAS assumptions. FL system characteristics include the average hardware capabilities of clients, the average network latency of clients, the number of participating clients, etc. Additionally, practitioners need to weigh which techniques to use to overcome their relevant set of challenges, since a technique for overcoming one challenge may inadvertently make it harder to overcome another.

However, finding suitable techniques is difficult. Prior surveys of the NAS-in-FL literature categorise NAS-in-FL methods as a whole rather than analyse the individual techniques they use. [19] categorises NAS-in-FL methods into offline versus online architecture search and single-objective versus multi-objective methods. [12] gives a high-level overview of the NAS-in-FL landscape at the time of its publication as part of a larger survey into combining NAS and Hyperparameter Optimisation. [8] investigates how multi-objective optimisation can be integrated into FL in general and includes only parts that discuss how this is done specifically for NAS-in-FL methods. Moreover, all of the surveys analyse only a small fraction of the entire NAS-in-FL literature. As a result, knowledge on challenges when using NAS in FL and the techniques for overcoming those challenges remains fragmented across several studies. A more sophisticated understanding of the challenges and remedying techniques for using NAS in FL is needed to allow practitioners to find techniques that could overcome the set of challenges relevant to their targeted set of FL system characteristics. Therefore, we ask the following research question:

What challenges arise when using NAS in FL, and what techniques overcome these challenges in the literature?

To tackle our research question, we conduct an extensive systematic literature review. We collect an initial set of relevant literature with a Scopus [6] search and extend it with a snowball search. We first use open coding to identify, from the literature, assumption discrepancies and resulting challenges, when using NAS in FL. Next, we extract FL system characteristics and use axial coding to code the influence of characteristics on the relevance of challenges. Then, we extract techniques through open coding and iteratively refine them to synthesise a coherent set of mutually exclusive and collectively

exhaustive techniques. Finally, we use axial coding to map techniques onto challenges they overcome.

Our review aims to synthesise techniques for using NAS in FL from the literature to make finding suitable techniques easy in practice. We make the following three contributions. First, by providing a consolidated overview of the challenges of using NAS in FL, practitioners can grasp which issues they can expect to encounter at a glance. Second, by describing how FL system characteristics influence the relevance of challenges, we enable practitioners to focus on the challenges relevant to them. Third, by extracting techniques for using NAS in FL and highlighting the challenges they address, we enable practitioners to compare existing techniques for using NAS in FL for their specific set of challenges.

In Chapter 2, we cover the background required for this thesis and related work. In Chapter 3, we describe the method with which we perform our literature review and give an overview of the included literature. In Chapter 4, we highlight the challenges with FedNAS and how they arise from different FL settings. In Chapter 5, we describe the adaptation techniques we synthesised and how they overcome challenges. In Chapter 6, we conduct a discussion about our work. Chapter 7 contains our conclusion.

2 Background and Related Work

NAS is contained entirely in the 5th step of the FL pipeline as described in [11].

2.1 Neural Architecture Search

When practitioners make use of Deep Learning, they usually design the architecture of a neural network manually before training it. This design process usually consists of tedious rounds of trial and error that is guided by heuristics and practitioners' experience. As a way to save time and make this process more reproducible, practitioners have attempted to automate it, which has culminated in a wide variety of methods . As a way to speed up and improve this process, practitioners invented methods for automatically designing the architecture of a neural network by searching for architectures that are suited to being trained well for a certain task

2.2 Federated Learning

2.3 Federated Neural Architecture Search

3 Method

[TODO: overhaul to include method for identifying challenges and FL settings] - targeted FL setting can be stated directly, indirectly or must be inferred from experimental setup

We use various techniques from [3] and lean on parts of the methodology of [10] to perform a systematic literature review.

1. **Literature Selection:** We follow the guidelines and flow diagrams provided by PRISMA 2020 [15] for inclusion and exclusion of papers and perform forward and backwards citation searching. We identify 59 papers that present FedNAS methods. Each paper contains one or more FedNAS methods.
2. **Unrefined Adaptation Technique Extraction:** Once the set of included papers is fixed, we perform open coding on each FedNAS method to extract unrefined adaptation techniques. Any modification to a NAS method that is explicitly motivated by the federated setting is treated coded as one unrefined adaptation technique.
3. **Adaptation Techniques Conceptualization:** We iteratively refine and merge unrefined adaptation techniques in an axial coding step to obtain a coherent set of adaptation techniques that are mutually exclusive and collectively exhaustive. Unrefined adaptation techniques with conceptually highly-similar mechanisms are merged into a single representative adaptation technique.
4. **Categorise Adaptation Techniques:** After merging, in a second axial coding step, we cluster adaptation techniques based on a) the FedNAS challenges they address and b) the conceptual similarity of their mechanisms. As a result, we obtain a taxonomy of adaptation techniques.
5. **Discuss FedNAS Challenges for Adaptation Techniques:** We discuss how each adaptation technique works towards, against, or does not affect overcoming each of FedNAS challenges.
6. **Produce Table Overviews:** Finally, we create two tables that practitioners can use to make decisions about FedNAS methods for their use case.

The first table contains a coded vector of effects over the FedNAS challenges for each *FedNAS method*. The effects per FedNAS method are the result of the effects of its adaptation techniques in aggregate. Practicioners can use this table to decide which FedNAS method suits their use case or if their use case would benefit from a new FedNAS method.

The second table table contains a coded vector of effects over the FedNAS challenges for each *adaptation technique* based on the prior discussion. If practitioners need to create new FedNAS methods, they can use this table to choose existing adaptation techniques relevant to the set of FedNAS challenges they need to address for their use case.

3.1 Reviewed Literature

add some overview and statistics about the literature

4 FedNAS Challenges

Assumption in Centralised Setting	Reality in FL Setting	Resulting FedNAS Challenge Description	FedNAS Challenge Name
Worker nodes' hardware is homogeneous.	Clients' hardware varies significantly. More pronounced in the cross-device setting than the cross-silo setting.	FedNAS methods need to be heterogeneity-aware; otherwise, the search will be delayed by low-end devices or leave high-end devices waiting in idle most of the time.	Hardware Heterogeneity
Worker nodes communicate over high-bandwidth, low-latency links.	Clients typically communicate with the central server over high-latency and low-bandwidth connections.	FedNAS must be communication-efficient; otherwise, exchanging model weights and architecture parameters becomes a bottleneck during the search process.	Limited Networking Capabilities
Worker nodes are high-end machines with powerful CPUs, GPUs, and large amounts of RAM.	Clients are edge devices with few computational resources.	The computational burden placed on clients by a NAS method needs to be adjusted such that the architecture search takes place within an acceptable time frame. This can involve splitting up computational work, usually done on single worker nodes, amongst multiple clients, making the implementation complex.	Limited Computational Resources
Training data is gathered at a central location.	Training data is distributed unevenly across clients	Since some clients have more data than others, FedNAS methods must ensure that these clients are not overrepresented in the searched architecture.	Unbalanced Client Data

The training data is drawn from the same distribution.	Each client draws training data from their own distribution.	FedNAS methods need to be robust with respect to non-i.i.d. data, which makes evaluating and ranking candidate architectures noisier than in the centralised setting.	Non-I.I.D. Training Data
Worker nodes are equally available.	Some clients are more frequently available than others.	FedNAS methods must prevent the search from being dominated by the most frequently available clients; otherwise, the resulting architecture will be biased towards them.	Variable Client Availability
All worker nodes participate in each iteration.	Only a subset of clients participates in each iteration of the search.	FedNAS methods must ensure that a realistic sample of the client population is represented in the architecture search and address high-variance performance estimates, as candidate architectures are evaluated on changing subsets of clients.	Client Participation
Worker nodes are inside the same trust domain.	All participating parties (i.e. the central server and clients) can consider another potentially malicious.	FedNAS methods must address attacks from participating parties, such as architecture parameter poisoning during the search process.	Security
Worker nodes consistently participate in the search process.	Clients can drop out of a communication round at any time, and completion times of epochs vary.	FedNAS methods need to handle client dropouts and stragglers, since interrupted or delayed evaluations of candidate architectures can slow down or destabilise the search process.	Client Reliability

Model accuracy is more important than model resource consumption in selecting architectures.	Model accuracy needs to be traded for model resource consumption in selecting architectures.	Models trained in FL tend to be deployed on the same resource-constrained clients they are trained on. Therefore FedNAS methods need to optimise the architecture of the trained model w.r.t. resource constraints.	Model Resource Constraints
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Table 4.1: Discrepancies between NAS in the centralised setting and the FL setting and the resulting FedNAS challenges. *Worker nodes* perform the architecture search in the centralised setting, whereas a *server* and *clients* perform the search in the FL setting. The challenges are based to some extent on previously identified challenges in the FL setting in general [14] [11] [4] [1].

5 Adaptation Techniques

5.1

6 Discussion

7 Conclusion

Abbreviations

List of Figures

List of Tables

- 4.1 Discrepancies between NAS in the centralised setting and the FL setting and the resulting FedNAS challenges. *Worker nodes* perform the architecture search in the centralised setting, whereas a *server* and *clients* perform the search in the FL setting. The challenges are based to some extent on previously identified challenges in the FL setting in general [14] [11] [4] [1]. 10

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